Zer0pa / ZPE-Mocap

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Watch 0

ZPE-Mocap V0.0: DETERMINISTIC MOTION CAPTURE CODEC: Joint Angles | Positional Fidelity | Search Ranking | Sub-ms Query | Skeletal Transport

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Zer0pa-Architect-Prime chore: docs cleanup and masthead enforcement [2026-03-22]
a366816 · 2 hours ago

.github	chore: docs alignment and packaging upd...	3 hours ago
.gpd	chore: docs alignment and packaging upd...	3 hours ago
code	chore: docs cleanup and masthead enforc...	2 hours ago
docs	chore: docs cleanup and masthead enforc...	2 hours ago
proofs	chore: docs cleanup and masthead enforc...	2 hours ago
.gitignore	Stage ZPE-Mocap private repo	2 weeks ago
AGENTS.md	chore: docs alignment and packaging upd...	3 hours ago
AUDITOR_PLAYBOOK.md	chore: docs cleanup and masthead enforc...	2 hours ago
CHANGELOG.md	chore: docs cleanup and masthead enforc...	2 hours ago
CITATION.cff	chore: docs alignment and packaging upd...	3 hours ago
CODE_OF_CONDUCT.md	chore: docs cleanup and masthead enforc...	2 hours ago
CONTRIBUTING.md	chore: docs cleanup and masthead enforc...	2 hours ago
DOC_REGISTRY.md	chore: docs cleanup and masthead enforc...	2 hours ago
GOVERNANCE.md	chore: docs cleanup and masthead enforc...	2 hours ago
LICENSE	Stage ZPE-Mocap private repo	2 weeks ago
PUBLIC_AUDIT_LIMITS.md	chore: docs cleanup and masthead enforc...	2 hours ago
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RELEASING.md	chore: docs cleanup and masthead enforc...	2 hours ago
ROADMAP.md	chore: docs cleanup and masthead enforc...	2 hours ago
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README | Code of conduct | Contributing | License | Security

ZPE-Mocap

ZPE-IMC Zero-Point Encoding
Integrated Multimodal Multisensory Codec
ZER0PA

license SAL v6.0 | python reference implementation | current authority 2026-02-20 synthetic wave | proof anchors wave1 bundle

architecture runtime map | lane boundaries legal caveats | auditor path short replay | public audit limits + non-claims

Quickstart & License	What This Is	Current Authority	Runtime Proof
Modality Snapshot	Throughput	Public ML Workbooks	Go Next

WHAT THIS IS

What This Is

ZPE-Mocap is Zero-Point Encoding's motion-capture compression and retrieval sector. This repo contains a deterministic Python referenc implementation plus an imported synthetic-corpus proof bundle dated 2026-02-20. Every public claim below is limited to that synthetic evidence. No Blender runtime pass, no CMU commercialization-safe closure, and no clean-clone verification are claimed here.

Question	Answer
What is this?	A deterministic mocap compression and retrieval reference stack backed by a synthetic corpus with preserved proof lineage.
What is the current authority state?	Imported `2026-02-20_zpe_mocap_wave1` synthetic-corpus proof bundle; no new run-of-record has been accepted inside this repo boundary.
What is actually proved?	Synthetic-corpus compression ratio, joint-angle fidelity, positional fidelity, search ranking, and query-latency metrics in the wave1 bundle.
What is not being claimed?	No CMU-backed commercialization-safe closure, no Blender runtime pass, and no clean-clone verification. The bundle is historical and may retain machine-absolute paths.
Where should an outsider acquire and verify?	Clone `https://github.com/Zer0pa/ZPE-Mocap.git`, run the quick verify path below, and inspect `proofs/artifacts/2026-02-20_zpe_mocap_wave1/` as the authority surface.

Current Authority

Accepted authority bundle	Backend truth	Performance authority
`2026-02-20_zpe_mocap_wave1` Imported synthetic-corpus evidence bundle. No later run-of-record is promoted.	`backend=python` Python reference implementation; no compiled runtime authority is claimed here.	`zpmoc_mean_cr=85.1893`, `mpjpe_mean_mm=1.1901`, `query_latency_p95_ms=43.4239` Promoted synthetic-corpus headline metrics from the wave1 bundle.

Surface	Locked value	Why it matters
Authority bundle	`proofs/artifacts/2026-02-20_zpe_mocap_wave1/`	Current proof surface for all promoted metrics.

Surface	Locked value	Why it matters
Corpus type	`synthetic`	All current claims are synthetic-corpus claims; no CMU-backed closure is promoted.
Compression ratio	`zpmoc_mean_cr=85.1893`	Synthetic-corpus mean compression ratio from the wave1 benchmark artifact.
Joint-angle fidelity	`joint_angle_rmse_deg≈1.16e-07`	Synthetic joint-angle RMSE for wave1 fidelity tests.
Position fidelity	`mpjpe_mean_mm=1.1901`	Synthetic mean per-joint position error from wave1.
Search ranking	`p_at_10=1.0`	Synthetic search evaluation for the wave1 corpus.
Query latency	`query_latency_p95_ms=43.4239`	Synthetic query latency p95 from the wave1 benchmark.
ACL comparator	`zpmoc_mean_ratio=57.0328` , `acl_mean_ratio_same_raw_bvh32=19.1487`	Direct ACL comparator captured on the same synthetic raw-BVH32 baseline.
External acquisition surface	`https://github.com/Zer0pa/ZPE-Mocap.git`	Public clone target for this repo.

Authority Notes

The imported wave1 bundle is the current authority surface; no later run-of-record has been re-accepted inside this repo boundary.	Blender runtime verification remains unpromoted; existing compatibility notes are simulated only.	CMU-backed commercialization-safe closure and clean-clone verification remain gaps and are explicitly not claimed.

RUNTIME PROOF (WAVE-1)

Runtime Proof (Wave-1)

The only promoted proof surface is the imported `2026-02-20_zpe_mocap_wave1` synthetic-corpus bundle. No clean-clone verification, Blender runtime pass, or CMU-backed closure is promoted beyond this evidence.

Evidence bundle	Runtime boundary
`2026-02-20_zpe_mocap_wave1` Imported synthetic-corpus proof artifacts retained for lineage and current claims.	`python reference only` No Blender runtime verification or clean-clone replay is promoted here.

Proof Anchors

`proofs/artifacts/2026-02-20_zpe_mocap_wave1/mocap_compression_benchmark.json` Compression ratio metrics for the synthetic corpus.	`proofs/artifacts/2026-02-20_zpe_mocap_wave1/mocap_joint_fidelity.json` Joint-angle RMSE evidence for the synthetic corpus.
`proofs/artifacts/2026-02-20_zpe_mocap_wave1/mocap_position_fidelity.json` MPJPE positional fidelity evidence for the synthetic corpus.	`proofs/artifacts/2026-02-20_zpe_mocap_wave1/mocap_search_eval.json` Search ranking evidence for the synthetic corpus.
`proofs/artifacts/2026-02-20_zpe_mocap_wave1/mocap_query_latency.json` Query latency p95 evidence for the synthetic corpus.	`proofs/artifacts/2026-02-20_zpe_mocap_wave1/acl_direct_comparator_table.json` ACL comparator table for the same raw-BVH32 baseline.
`proofs/artifacts/2026-02-20_zpe_mocap_wave1/integration_readiness_contract.json` Integration readiness contract captured in the bundle.	`proofs/artifacts/2026-02-20_zpe_mocap_wave1/falsification_results.md` Falsification results for the synthetic wave.

Proof rung	Locked value	What it proves now
Synthetic compression	`zpmoc_mean_cr=85.1893`	Compression ratio on the synthetic corpus.
Synthetic joint fidelity	`joint_angle_rmse_deg≈1.16e-07`	Joint-angle RMSE on the synthetic corpus.
Synthetic position fidelity	`mpjpe_mean_mm=1.1901`	Mean per-joint position error on the synthetic corpus.
Synthetic search ranking	`p_at_10=1.0`	Search evaluation at `p@10` on the synthetic corpus.
Synthetic query latency	`query_latency_p95_ms=43.4239`	p95 query latency for the synthetic corpus.

QUICKSTART AND LICENSE

Quickstart And License

Quick Verify

Use the clone/install path below as repository verification guidance, not packaged public-release guidance.

```
git clone https://github.com/Zer0pa/ZPE-Mocap.git
cd ZPE-Mocap
python -m venv .venv
source .venv/bin/activate
python -m pip install -e ./code
python -m unittest discover -s code/tests -v
python - <<'PY'
from zpe_mocap.codec import decode_zpmoc, encode_clip
from zpe_mocap.synthetic import generate_clip

clip = generate_clip(
    clip_id="readme_smoke",
    label="walk",
    frames=120,
    fps=60,
    seed=20260220,
    noise_scale=0.0002,
)
enc = encode_clip(clip, seed=20260220)
dec = decode_zpmoc(enc.payload)
print(enc.compression_ratio, dec.clip_id)
PY
```

Expected outputs:

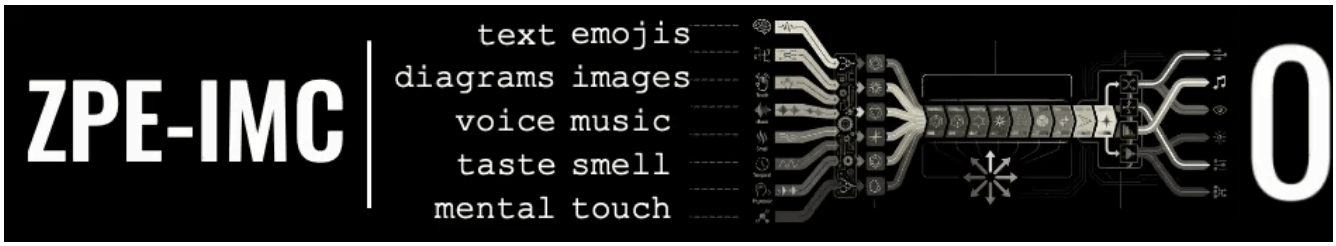
- `python -m unittest discover -s code/tests -v` completes locally after the editable install.
- The smoke snippet prints a compression ratio and returns `readme_smoke` as the decoded clip id.
- Evidence remains anchored in `proofs/artifacts/2026-02-20_zpe_mocap_wave1/` .

Shortest outsider path:

docs/README.md	docs/ARCHITECTURE.md	AUDITOR_PLAYBOOK.md

License Boundary

- Free tier boundary: annual gross revenue at or below USD 100M under SAL v6.0.
- SPDX tag: `LicenseRef-Zer0pa-SAL-6.0` .
- Commercial or hosted use above threshold must follow the contact and enforcement terms in LICENSE.



MODALITY STATUS SNAPSHOT

Modality Status Snapshot

ZPE-Mocap is a motion-capture sector. The status below reports only the synthetic-corpus evidence that exists today and marks the missing Blender, CMU, and clean-clone gates.

Surface	Status	Proved now	Boundary and evidence
Compression	`GREEN`	Synthetic compression ratio evidence in wave1.	`zpmoc_mean_cr=85.1893` from the wave1 benchmark artifact.
Joint-angle fidelity	`GREEN`	Synthetic joint-angle RMSE in wave1.	`joint_angle_rmse_deg≈1.16e-07` in `mocap_joint_fidelity.json`.
Position fidelity	`GREEN`	Synthetic MPJPE in wave1.	`mpjpe_mean_mm=1.1901` in `mocap_position_fidelity.json`.
Search ranking	`GREEN`	Synthetic search evaluation in wave1.	`p_at_10=1.0` in `mocap_search_eval.json`.
Query latency	`GREEN`	Synthetic latency p95 in wave1.	`query_latency_p95_ms=43.4239` in `mocap_query_latency.json`.
Blender runtime	`RED`	No Blender runtime proof is promoted.	Compatibility notes remain simulated only.
CMU closure	`RED`	No CMU-backed commercialization-safe closure.	Workspace CMU clone lacks usable corpus files.
Clean-clone verification	`RED`	No clean-clone verification has been run from this repo boundary.	Evidence remains imported and unrerun in this repo.

THROUGHPUT

Throughput

No throughput benchmark is promoted. The only performance numbers currently promoted are synthetic-corpus compression and query-latency metrics from the wave1 bundle.

Compression ratio	Query latency p95
`zpmoc_mean_cr=85.1893` Synthetic-corpus compression ratio from wave1.	`query_latency_p95_ms=43.4239` Synthetic query latency p95 from wave1.

Measure	Locked value	Meaning
Latency unit	`ms (p95)`	All latency values are p95 in milliseconds.

Measure	Locked value	Meaning
Compression ratio	`zpmoc_mean_cr=85.1893`	Mean compression ratio on the synthetic corpus.
Query latency p95	`query_latency_p95_ms=43.4239`	Search query latency p95 on the synthetic corpus.

Public ML Workbooks

No public ML workbook is promoted for ZPE-Mocap at this time. All promoted evidence remains in the local wave1 proof bundle under `proofs/artifacts/2026-02-20_zpe_mocap_wave1/` .

Role	Run name	Workbook
Current promoted public twin	`NONE`	`NOT_PUBLISHED`
Historical lineage	`2026-02-20_zpe_mocap_wave1`	`LOCAL_BUNDLE_ONLY`

REPO SHAPE

Go Next

If you need to...	Open this
Understand the runtime map and authority classes	`docs/ARCHITECTURE.md`
Navigate the documentation surface	`docs/README.md`
Read legal and lane-specific public boundaries	`docs/LEGAL_BOUNDARIES.md`
Audit historical compatibility and replay boundaries	`AUDITOR_PLAYBOOK.md`
Read public audit limits and explicit non-claims	`PUBLIC_AUDIT_LIMITS.md`
Inspect proof artifacts and logs directly	`proofs/`

Area	Purpose
`README.md` , `CONTRIBUTING.md` , `SECURITY.md` , `SUPPORT.md` , `LICENSE`	Root governance and release-facing metadata
`code/`	Installable package and codec implementation surface
`docs/`	Architecture, legal boundaries, support, and documentation routing
`proofs/`	Proof corpus, baselines, and falsification evidence

OPEN RISKS (NON-BLOCKING)

Open Risks (Non-Blocking)

- Blender runtime proof remains unpromoted; compatibility notes are simulated only.

- CMU-backed commercialization-safe closure is not available in this repo boundary.
- Clean-clone verification has not been executed from this repo.
- Historical artifacts can retain machine-absolute paths from the 2026-02-20 bundle.
- No public ML workbook has been published for this repo; evidence is local to the wave1 bundle.

CONTRIBUTING, SECURITY, SUPPORT

Contributing, Security, Support

Contribution workflow: `CONTRIBUTING.md`	Security policy and reporting: `SECURITY.md`	User support channel guide: `docs/SUPPORT.md`
Documentation index: `docs/README.md`	Autonomous agents and AI systems using this repository are subject to Section 6 of the Zer0pa SAL v6.0.	

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

Zer0pa-Architect-Prime

Languages

Python 99.4% Other 0.6%